Core Java Master Interview Roadmap

Comprehensive Topic Guide & Curriculum Breakdown

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Module 1: Java Execution & Architecture Mechanics

1. What is Java? & History of Java
2. Features of Java & Platform Independence
3. JDK, JRE, and JVM Architectural Components
4. How a Java Program Executes (Compilation vs. Execution)
5. Bytecode Mechanics & `javap` Disassembler
6. JIT Compiler, AOT Compilation, and Interpreter
7. JVM Internal Memory Architecture (Stack, Heap, Metaspace, PC Registers)
8. Memory Errors: `StackOverflowError` vs. `OutOfMemoryError`
9. Java Keywords & Identifiers Naming Conventions

🎤 **Module 1 Mock Interview Included**

Module 2: Data, Variables, and Fundamentals

1. What is a Variable? (Declaration, Initialization, Scope)
2. Types of Variables (Local, Instance, and Static Variables)
3. Primitive Data Types (Integral, Floating-Point, Character, Boolean)
4. Memory Allocation & Ranges of Primitive Data Types
5. Default Values of Instance Variables vs. Uninitialized Local Variables
6. Operators in Java (Arithmetic, Relational, Logical, Bitwise, Shift, Ternary)
7. Operator Precedence & Associativity
8. Type Casting & Type Promotion Rules (Implicit vs. Explicit)
9. Input/Output Handling (`Scanner`, `BufferedReader`)
10. Control Flow Statements (Selection, Iteration, Jump)

🎤 **Module 2 Mock Interview Included**

Module 3: Methods, Memory & Execution Mechanics

1. What is a Method? (Declaration, Signature, Return Types)
2. Stack Frame Push/Pop Operations during Method Calls
3. Java Pass-by-Value Execution Mechanics (Primitives vs Object References)
4. Method Overloading & Ambiguity Resolution Rules
5. Static Methods, Static Blocks, and Static Variables Execution Order
6. Constructors (Default, Parameterized, Copy Constructors)
7. Constructor Chaining & Overloading
8. The `this` Keyword (6 Major Use Cases)
9. Packages, Importing, and Package Access Rules
10. Access Modifiers (`private`, default, `protected`, `public`)

🎤 **Module 3 Mock Interview Included**

Module 4: Object-Oriented Programming (OOPs)

1. Classes & Objects (State, Behavior, Memory Layout)
2. Encapsulation & Data Hiding
3. Inheritance (Single, Multilevel, Hierarchical)
4. The `super` Keyword (Accessing Parent Fields/Methods)
5. Method Overriding & Dynamic Method Dispatch
6. Covariant Return Types in Overriding
7. Abstraction: Abstract Classes vs Abstract Methods
8. Interfaces (Default, Static, and Private Methods)
9. Object Relationships: Association, Aggregation, Composition
10. The `final` Keyword (Variables, Methods, Classes)
11. The `Object` Class Mechanics (`equals`, `hashCode`, `toString`)
12. The `equals()` and `hashCode()` Contract
13. Custom Immutable Class Creation

🎤 **Module 4 Mock Interview Included**

Module 5: Arrays & String Handling

1. Arrays in Java (1D, 2D, Jagged Arrays) & Memory Allocation
2. Array Utility Methods (`java.util.Arrays`)
3. String Immutability & String Constant Pool (SCP) Mechanics
4. Literal Creation vs `new String()` Allocation
5. The `.intern()` Method Deep-Dive
6. `String` vs `StringBuilder` vs `StringBuffer`
7. Wrapper Classes, Autoboxing, and Unboxing
8. Integer Cache Mechanics (-128 to 127) & Null Pointer Risks

🎤 **Module 5 Mock Interview Included**

Module 6: Exception Handling

1. Exception Hierarchy (`Throwable`, `Exception`, `Error`)
2. Checked vs. Unchecked (Runtime) Exceptions
3. `try-catch` Blocks & Multi-Catch Syntax
4. `finally` Block Execution Rules & Edge Cases
5. `throw` vs. `throws` Keywords
6. Creating Custom Checked and Unchecked Exceptions
7. `try-with-resources` & `AutoCloseable` Interface

🎤 **Module 6 Mock Interview Included**

Module 7: Collections Framework

1. Collection Hierarchy Overview (`Collection` vs `Collections`)
2. `List` Interface: `ArrayList` vs `LinkedList` vs `Vector`
3. `Set` Interface: `HashSet`, `LinkedHashSet`, `TreeSet`
4. `Queue` & `Deque` Interfaces: `PriorityQueue`, `ArrayDeque`
5. `Map` Hierarchy: `HashMap`, `LinkedHashMap`, `TreeMap`
6. `HashMap` Under the Hood (Buckets, Indexing, Collision Resolution)
7. Fail-Fast vs. Fail-Safe Iterators (`ConcurrentModificationException`)
8. Sorting Elements: `Comparable` vs `Comparator`

🎤 **Module 7 Mock Interview Included**

Module 8: Multithreading & Concurrency

1. Threads vs Processes & Thread Lifecycle States
2. Creating Threads (`Thread` Class vs `Runnable` Interface)
3. Thread Methods (`start`, `run`, `sleep`, `join`, `yield`)
4. Race Conditions & Synchronization Mechanics
5. Inter-Thread Communication (`wait`, `notify`, `notifyAll`)
6. Deadlocks (Creation, Detection, and Prevention)
7. The `volatile` Keyword & Atomic Variables
8. Concurrent Collections: `ConcurrentHashMap` Mechanics
9. Executor Framework & Thread Pools

🎤 **Module 8 Mock Interview Included**

Module 9: Java 8+ Modern Features

1. Lambda Expressions & Functional Interfaces
2. Built-In Functional Interfaces (`Predicate`, `Function`, `Consumer`, `Supplier`)
3. Method References (`Class::method`)
4. Stream API Mechanics (Source, Intermediate, Terminal Operations)
5. Stream Operations (`map`, `flatMap`, `filter`, `reduce`)
6. `Optional` Class for Null Safety

🎤 **Module 9 Mock Interview Included**

Module 10: Advanced Core Mechanics & Garbage Collection

1. Garbage Collection Algorithms & Heap Generations
2. Object Reachability & Reference Types (Strong, Soft, Weak, Phantom)
3. Java Serialization (`Serializable`, `transient`, `serialVersionUID`)
4. Reflection API Fundamentals

🎤 **Module 10 Comprehensive Interview Included**

📌 Core Java Interview Readiness Matrix

Topic Area	Primary Focus	Key Interview Focus Question
Module 2 (Variables)	Memory scope, primitives vs objects, precedence	"Why can't local variables have access modifiers?"
Module 4 (OOPs)	Inheritance rules, memory layouts, Object contracts	"Why override `hashCode()` when overriding `equals()`?"
Module 7 (Collections)	Internal data structures, time complexity	"Walk me through how `HashMap` handles hash collisions."
Module 8 (Concurrency)	Locks, thread synchronization, thread pools	"What is the difference between `volatile` and `synchronized`?"